GOVT. POLYTECHNIC
BALASORE

DEPARTMENT OF ELECTRICAL ENGINEERING

CERTIFICATE

NAME : Saroj Parida

REGISTRATION NO : F241100 2014

BRANCH : Electrical

SECTION : B2

SUBJECT : Programming for problem Solving

This is certified to be bonafide work of the student in the Programming for problem solving laboratory (PR.2) during the academic year 2025 / 26.

No. of practicals certified **10** out of **10** in the subject of Programming for problem solving laboratory(PR.2).

Date: 26/11/25 **Teacher In-Charge**

Displaying College Name Multiple Times:

```
#include <stdio.h>
int main(){
    int i;
    for(i=1; k=20; i++){
        printf("%d. Government polytechnic
               Balasore\n", i);
    }
    return 0;
}
```

Submitted by:

Name: Saroj Parida
Reg.No: F241100914
Branch: Electrical
Group: B2
Sem: 3rd

Arrays:-

12. Arrays are collections of similar data elements stored contiguously in memory.

13. One-dimensional Array: Stores a list of values (e.g. Marks of 10 students)

14. Two-dimensional Arrays: Stores a table or matrix of values (e.g. matrix[25][25]).

7. Operators perform specific actions:

- Arithmetic: +, -, *, /, % (modulus)
- Relational: <, >, <=, >=, ==, != (not)
- Logical: && (AND), || (OR), ! (NOT)

Input/Output (I/O) operations:

8. Input Function (scanf()): Reads data from the keyboard. It requires a format specifier and the address(&) of the variable.

9. Output Function (printf()): Display data or message to the console. It uses format specifiers to output variable values.

Control Structures (Decision Making and Looping):

10. Decision Making structures allow conditional execution of code.
    - if Statement: Executes code if a condition is true.
    - if-else statement: Executes one block if true, another if false.
    - Switch Statement: Used for multi-way branching based on a single variable's value.

11. Looping Structures execute code repeatedly.
    - for loop: Used when the number of iterations is generally known.
    - while loop: Repeats as long as a condition remains true. (pre-test loop).

Programming for Problem Solving

Introduction & Basic Structure:

1. C Programming is a general-purpose, procedural language developed by Dennis Ritchie.

2. It is widely used for system programming and as a foundation for many other languages.

3. The basic structure of a C programme includes:
   - Preprocessor Directives: Instructions starting with # (e.g. #include <stdio.h>)
   - Main Function: The starting point of execution (int main() { ... })

   Data types, variables, and operators:

4. Declarations: Defining variables and functions.
   Executable Statements: The instructions that define the program's logic.

5. Constants: are fixed values that do not change (e.g. Integer 10, Character 'B').

6. Variables are names given to storage areas whose values can change during execution.
   Data Types define the type and size of data:
   - int: for whole numbers.
   - float/double: for decimal numbers.
   - char: for single characters.

INDEX

Sl. No	Name of the Experiment	Page No.	Date of experiment	Date of Submission	Remarks
01	Programming for Problem Solving:	1-5	09/09/25	26/09/25	
02	Displaying College Name Multiple times	6	16/09/25	23/09/25	
03	Sum of Even Number(1-10)	7	23/09/25	06/10/25	
04	Finding Smallest Number in an Array	8	25/09/25	06/10/25	
05	Sorting Array Elements in Ascending order	9	06/10/25	13/10/25	
06	Calculate Factorial of a number	10	13/10/25	14/10/25	
07	Calculate multiplication of two 2x2 matrices	10	14/11/25	24/10/25	
08	Recursion	10	19/10/25	03/11/25	
09	MATLAB Programming - (Addition, Subtraction, Multiplication)	13-13	28/10/25	03/11/25	
10	Plotting a Sinusoidal wave figure	14	09/11/25	10/11/25	
11	Solving linear Equation using matrix Division	15	11/11/25	11/11/25	
12	MATLAB Simulink: Modeling a simple DC Motor speed Control	16-18	11/11/25	25/11/25	